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EDUCATION

Communication University of China (CUC)			Beijing
<i>M.S. in Communication and Information Systems</i>	GPA: 3.85/4.00 (Top 5%)	Sep. 2021 – Jun. 2024	China
Communication University of China (CUC)			Beijing
<i>B.Eng. in Communication Engineering</i>	GPA: 3.60/4.00 (Top 10%)	Sep. 2017 – Sep. 2021	China

Honors & Awards: Two S-level Performance Ratings (2025); Selected Outstanding Campus Hire (Algorithm Track), Department Technical Breakthrough Award, Algorithm Supernova, and Outstanding Individual (Local Services), Kuaishou; Tomorrow of Stars, Microsoft Research Asia (Highly Selective); National First Prize, Student Innovation & Entrepreneurship Program; CCTV Scholarship; First-Class Scholarship, Outstanding Student & Student Leader, and Outstanding Graduate, CUC

PROFESSIONAL SUMMARY

- Recommendation foundation-model engineer specializing in next-generation recommender systems, recommendation foundation models, and large-scale ranking optimization. Currently serving as the dotted-line technical lead for next-generation recommendation exploration, owning direction planning and project execution while leading multiple team members across Foundation Recommendation Models, Agentic Recommendation, and scenario-aware multimodal content foundations. Earned two promotions within two years and two S-level performance ratings in 2025; recognized as a Selected Outstanding Campus Hire (Algorithm Track), Department Technical Breakthrough Award recipient, and Algorithm Supernova, with multiple Outstanding LR recognitions. First author of multiple AAAI and SIGIR papers and a core contributor to the OneReason technical report; mentored four interns, two of whom received full-time offers, earned Outstanding LR ratings, and produced publications.
- Research Experience:** State Key Laboratory of Media Convergence and Communication (**MCC Lab**); Social Computing Group, Microsoft Research Asia (**MSRA**)
- Technical Focus:** Foundation Recommendation Models, Agentic Recommendation, Generative Recommendation, Multimodal Content Understanding and Domain Foundation Models, Responsible AI, LLM Safety & Societal Impact
- Technical Skills:** *Programming Languages:* Python (with PyTorch and TensorFlow), Java, C++, and MATLAB; *Systems and Tools:* Experienced with Linux, Apache Spark, Apache Hive, and Microsoft Office.

PROFESSIONAL EXPERIENCE

Kuaishou Technology	<i>LLM & Recommender Systems Algorithm Engineer</i>	Jul. 2024 – Present
Next-Generation Recommendation Systems & Technical Exploration (Dotted-Line Technical Lead): Own the technical roadmap, workstream decomposition, and project execution; lead multiple team members across Foundation Recommendation Models, Agentic Recommendation, and scenario-aware multimodal content foundations. Established capabilities spanning foundation-model training, latent reasoning, user memory, policy planning, tool-orchestrated retrieval, and unified ranking, driving production deployments, research publications, and technical reports.		
➤ Recommendation Foundation Model — OneReason (Technical Lead): Built OneReason on open-source LLMs including Qwen3.5-9B with a three-stage Pre-training, SFT, and Post-training pipeline; contributed deeply to SFT design and foundation-model training, and led production deployment. Proposed a fast-slow online recommendation architecture and domain-adaptation strategy for local-services advertising, delivering +4% posterior expected ad spend in online A/B tests. For reasoning, contributed to itemic-token semantic alignment, three-level cognition-enhanced CoT, and specialize-then-unify RL under a perception-cognition framework, enabling the Think-then-Answer paradigm in recommendation foundation models. For efficient reasoning, led WhisperRec, compressing multi-view adaptive CoT into latent reasoning tokens through three-stage alignment and curriculum post-training; improved SID@64 by 17.44% and 9.33% over explicit-CoT and No-Think variants, respectively, with over 10× online inference throughput , and delivered an additional +5% posterior expected ad spend in local-services advertising A/B tests. OneReason WhisperRec		
• Agentic Recommendation (Technical Lead): Lead multiple team members in replacing the conventional retrieval –		

pre-ranking – ranking funnel with a transparent, self-evolving Agentic Recommendation architecture built on a Foundation Recommendation Model. User Memory continuously captures long-term and real-time interests; the Planner generates recommendation policies, selects retrieval paths, and orchestrates tools; the Actor autonomously invokes retrieval routes and allocates weights and quotas across channels. The tool layer supports distilled embedding retrieval, direct Foundation Model retrieval, and distilled UserCF/ItemCF, followed by unified Foundation Model reasoning and ranking, forming an end-to-end Memory – Planning – Tool Use – Actor – Ranking loop. The system is under continuous iteration and has entered online A/B testing.

- **Scenario-Aware Multimodal Content Understanding & Representation Foundation (Core Contributor):** Independently built from scratch a production LLM4Rec stack spanning SFT, post-training, inference, and online serving; deployed semantic-ID enhancements into ranking across three production launches, delivering **+8% GMV** and **+6% orders**. Built a domain content-understanding foundation for local-services products and videos through joint QA and retrieval training, unifying generative comprehension with embedding representations. Designed compression-based prompts and learnable Chorus Tokens to aggregate cross-modal semantics, and combined generative and contrastive objectives through compression-aware attention to improve retrieval representations while preserving QA and generation capabilities. Quantized the model into a unified semantic-ID system that supplies content semantics and training signals to u2i retrieval, OneRec, ranking models, and the Foundation Recommendation Model. Further explored agent-based autonomous training for domain foundation models through template, ad-understanding, user-interaction, complex-reasoning, and quality-evaluation agents; the resulting AdLoop work is under review at AAAI 2027.
 - **Spatial Semantic Representation — LGSID Series (Core Contributor):** Proposed LGSID, the first geo-aware semantic representation framework for near-field recommendation, addressing the lack of spatial awareness in traditional ID-based ranking and the content-heavy, geography-light bias in structured multimodal prompts. Designed RL-based post-training algorithms from G-DPO to G-GSPO and a hierarchical geo-aware tokenization model (HGIT) to explicitly inject relative location and distance into the representation space; evolved LGSID into LGSID++, with three production launches delivering **+10% GMV** and **+8% orders**. Further proposed LGRID, a white-box generative disentanglement framework that moves residual quantization into the LLM generation space, bridging generative and representation models and delivering an additional **+5.8% GMV** and **+4% orders**. The first-author work was accepted to the AAAI 2026 Research Track, with LGRID under review at KDD 2027. [Paper Link](#) [Code Link](#) [Kuaishou Tech Paper Weekly](#)
 - **Ranking Model Optimization:** Optimized large-scale ranking for Kuaishou local-services advertising along two tracks—LLM-based architecture scaling and ID-based representation/spatial modeling—to improve computational efficiency and interest matching.
- **LLM-Based Ranking:** Explored next-generation ranking architectures inspired by LLM scalability and generative recommendation paradigms.
- **Large-Scale Ranking Model — ScaleLLM3 (Core Contributor):** Redesigned the ranking architecture to address complex discrete feature interactions, large but sparsely trained embedding tables, and inefficient long-sequence modeling. Combined stackable RankMixer/TokenMixer-Large feature-interaction modules with OneTrans/HSTU-style decoder-only sequence modeling, improving GPU utilization by **22pp**, FLOPs/sample by **5×**, and CTCVR AUC by **0.42pp**; currently advancing online replacement through cold-start optimization and model distillation.
 - **End-to-End Generative Ranking Model (University Collaboration Lead):** Addressed the information loss, leakage, and objective misalignment inherent in multi-stage recommendation funnels by building a unified candidate-generation and scoring model. In generation, introduced thinking tokens and supervised, controllable reasoning labels to model the spatial constraints and temporal periodicity of local-services behavior, improving Hit@5 by **0.17pp** over the generative baseline. In scoring, built a decoder-only ranker inspired by OneRanker and an online pipeline that distills production ranking scores as supervision to learn relative ordering among generated candidates. The scoring model is under training and validation, with online experiments planned next.
 - **ID-Based Ranking:** Improved spatial awareness and interest matching under the strong geographic constraints and irreducible sample sparsity of near-field distribution through model architecture and spatial feature upgrades.

- **Dense-Layer Optimization (Core Contributor):** Addressed irreducible sample sparsity caused by strict geographic constraints in local services, which weakens representation learning in both dense and sparse layers. Designed a self-supervised representation enhancement framework with contrastive learning and a feature-disentangled GMV prediction tower, increasing the importance of GMV and conversion features without degrading baseline prediction quality and delivering **+6% GMV** and **+4% orders**.
- **Sparse-Layer Optimization — ReST (Core Contributor):** Proposed ReST, a plug-and-play ID representation enhancement network for sparse, geographically constrained scenarios, consisting of Meta ID warm-start and spatially constrained ID enhancement modules. Introduced a domain-aware hybrid negative-sampling strategy to improve ID embedding quality, delivering **+1.2% overall GMV** and **+5% attributed GMV** while improving long-tail high-quality POI matching; the first-author paper was accepted to SIGIR 2026. [Paper Link](#)
- **Spatial Preference Alignment (Core Contributor):** Modeled periodic user interests across time and location through interactions over user behavior $\times$ user segmentation $\times$ spatiotemporal factors, delivering **+4% GMV** and **+2% orders**. Further identified surrounding supply as a key driver of local-services intent and addressed the limited context awareness of point-wise ranking by building GeoHash indexing infrastructure and a cross-attention mechanism over user demand, spatial constraints, and supply context, delivering **+10% GMV** and **+5% orders**.

Xiaohongshu (RedNote)

Algorithm Engineer Intern.

Aug. 2022 – Aug. 2023

- **Home Feed Ranking (New User Model Optimization):** Led iteration and optimization of homepage ranking models for new users, focusing on cold-start modeling and interest representation under sparse behavior signals.
 - **Cold Start Modeling:** Improved robustness and generalization of new-user models via warm-start initialization from mature user models, combined with ensemble learning and population-aware modeling (POSO); achieved +0.141% next-day retention. Developed zero-shot learning framework to generate missing behavioral features by transferring patterns from existing users, constructing generalized user interest representations and delivering +0.467% retention; further explored generative modeling approaches (Prompt Learning, VAE) to augment user behavior and learn latent ID representations, mitigating feature sparsity. Additionally incorporated graph-based modeling over user interactions (click/follow) and attributes to capture social structures, enhancing representation via similar-user retrieval and clustering.
 - **Interest Modeling:** Captured both short-term and long-term user interests using DIN-based architectures, achieving +0.121% retention; further explored multi-interest modeling approaches (e.g., MIND, ComiRec) to improve representation diversity and accuracy.
 - **Multi-task Loss Optimization:** Addressed distributional differences in new-user data by applying adaptive multi-task loss balancing methods (GradNorm, UWL, DTP), dynamically reweighting objectives to improve training stability.
- **Live Streaming In-feed Recommendation:** Led end-to-end optimization of live streaming in-feed recommendation systems, covering model improvements and production-level system development.
 - **Multi-objective Optimization:** Introduced deep engagement signals into ranking models and improved multi-task learning via enhanced MMoE architecture to mitigate objective conflicts, achieving +9.6% 10s view PV, +9.67% watch time, and +16.6% follows in online experiments.
 - **Cross-domain Transfer Learning:** Designed domain-adaptive transfer learning framework to leverage features from the main feed into live streaming scenarios while reducing domain bias via adaptive modules; achieved +0.3% AUC offline and +7.86% 10s view PV, +5.11% watch time, +8.71% follow UV online.
 - **Model Architecture Enhancement:** Upgraded ranking models with SeNet++ (Squeeze-and-Excitation Networks) for feature selection and improved Transformer-based interest modeling, achieving +0.3% AUC (feature modeling) and +0.1% AUC (interest modeling).

Multi-scenario Modeling: Explored joint modeling of live streaming in-feed and out-feed scenarios to reduce system maintenance cost and improve cross-scenario generalization, enabling unified modeling across recommendation surfaces.

RESEARCH EXPERIENCE

Microsoft Research Asia, Social Computing Group

Research Intern.

Sep. 2023 – May 2024

- **AI Safety & Societal Impact (LLM Research):** Conducted research on large language model safety and societal impacts, focused on understanding and improving robustness, interpretability, and controllability of LLMs.
 - **Adversarial Attacks :** Investigated self-reflection and self-revision capabilities of LLMs to design self-modifying adversarial inputs that induce model failure, revealing vulnerabilities in instruction-following and alignment mechanisms.
 - **Mechanistic Analysis of LLM Jailbreaks:** Analyzed internal neuron activation patterns in white box LLM before and after adversarial attacks to understand the mechanisms behind jailbreak success, providing insights into model failure modes at the representation level.
 - **Machine Unlearning for LLMs:** Constructed knowledge-graph-based benign and harmful triplets and applied unlearning to remove harmful information from model parameters; improved model safety while preserving utility. [Paper Link](#)
 - **LLM4Rec Security:** Studied robustness issues in LLM-based RecSys, demonstrating their brittleness to prompt perturbations and content manipulation; designed adversarial strategies that degrade recommendation performance.
- **LLM-based News Recommendation (MSN News / UniUM):** Participated in the design of UniUM ranking and retrieval systems for Microsoft MSN News, focusing on improving user modeling and retrieval quality through large language models.
 - **Dwell-Time-aware News Recommendation:** Proposed a user dwell-time-aware ranking model that incorporates dwell duration as a complementary signal to clicks, mitigating false-positive feedback signals via adaptive parameter fusion; achieved +0.44% and +0.32% AUC offline improvement and was fully deployed in production. [Paper Link](#)
 - **LLM-based I2I Retrieval Optimization:** Addressed noise and redundancy in multi-channel item-to-item retrieval by introducing GPT-based semantic filtering; replaced traditional I2I retrieval with LLM-driven semantic retrieval to enhance content quality, leveraging LLM semantic understanding to surface high-quality but previously missed items.

CUC, State Key Laboratory of Media Convergence and Communication

Research Student.

Sep. 2021 – Jul. 2024

- **News Recommendation Research:** Conducted research and system development for news recommendation, focusing on key challenges including explainability, cold-start, and diversity.
 - **Explainable News Recommendation:** Proposed a heterogeneous graph reasoning framework for news recommendation that leverages reinforcement learning to construct personalized user and news subgraphs, effectively filtering noisy click signals and identifying true user interests. Designed a multi-task learning framework integrated with contrastive learning to address sparse reward signals and slow convergence in reinforcement learning, improving training efficiency and recommendation interpretability. [Paper Link](#)
 - **Cold-Start for Users & News:** Introduced a self-supervised contrastive learning-enhanced cold-start representation framework, combining unsupervised and supervised learning to generate virtual representations for cold users and cold items. Designed a personalized gating mechanism to effectively integrate raw features with learned representations, significantly alleviating cold-start challenges in news recommendation systems. [Paper Link](#)
 - **Relation-aware News Understanding & User Modeling:** Constructed a multi-view news graph to model inter-news relationships and exploit collaborative signals among articles, enhancing central news representations through relational context. Further developed relation-aware user behavior modeling to capture interpretable user interest representations at the interaction level, leading to improved overall recommendation performance. [Paper Link](#)

PUBLICATIONS [Google Scholar](#)

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- [1] Jiang H, Du P, Yao P, et al. WhisperRec: Latent Reasoning for Efficient Foundation Recommendation Models. (ArXiv)
 - [2] OneRec Team, Jiang H, et al. OneReason Technical Report. (ArXiv)
 - [3] Jiang H, et al. AdLoop: An Agentic Feedback Loop for Domain-Specific LLMs Post-Training in Local-Life Service Advertising (AAAI 2027, Under Review)
 - [4] Jiang H, Wang Guoquan, Zhou Guorui et al. LLM-Aligned Geographic Item Tokenization for Local-Life Recommendation. (AAAI2026)

[5] **Jiang H**, Long Zhang, Zhou Guorui et al. Interpretable Representation via LLM-Driven Generative Disentanglement for Local Services Recommendation. (**KDD2027, Under Review**)

[6] **Jiang H**, Wang Guoquan, Zhou Guorui et al. ReST: A Plug-and-Play Representation Enhancement Framework for Spatially-Constrained Long-tail Item Recommendation. (**SIGIR2026**)

[7] **Jiang H**, Li C, An M. Time Matters: Enhancing Pre-trained News Recommendation Models with Robust User Dwell Time Injection[J]. arXiv preprint arXiv:2405.12486, 2024. (**Arxiv**)

[8] Mai, P. *, **Jiang, H***, Yan, R., Yang, Y., Huang, Z., and Pang, Y. Knowledge graph unlearning to defend language model against jailbreak attack. (**ICLR2024**)

[9] **Jiang H**, Li C, Cai J, et al. RCENR: A Reinforced and Contrastive Heterogeneous Network Reasoning Model for Explainable News Recommendation[C]//Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval. 2023: 1710-1720. (**SIGIR2023**)

[10] **Jiang H**, Li C, Cai J, et al. Self-supervised Contrastive Enhancement with Symmetric Few-shot Learning Towers for Cold-start News Recommendation[C]//Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. 2023: 945-954. (**CIKM2023**)

[11] **Jiang H**, Li C, Cai J, et al. MNN4Rec: A relation-aware approach based on multi-view news network for news recommendation[J]. Journal of Information Science. 2023: 01655515231182072. (**JIS, SCI-Q3**)

[12] **Jiang H**, Zhang L. Design of UWB Microstrip Antenna Based on A New Optimization Algorithm[C]//2022 Asia-Pacific International Symposium on Electromagnetic Compatibility. IEEE, 2022: 231-233. (**APEMC, EI**)

TEACHING EXPERIENCE

CUC, School of Information and Communication Engineering	<i>Teaching Assistant for C++ Course</i>	Spring-2022
• Collaborated with the lead instructor in delivering lectures and facilitating classroom discussions. • Addressed and clarified students' queries regarding programming concepts and assignments during class. • Assisted in grading assignments and provided constructive feedback to promote students' understanding and progress.		